

Teaching target

What must be built and repaired?

Capability

Track v_y through flight; distinguish velocity/acceleration; recognise apex; connect story, table, graph and equation.

Difficulty

Learners confuse negative acceleration with downward motion and apex with zero acceleration.

Meaning before formula

Build the physical idea first.

Phenomenon

A projectile may move upward while acceleration points downward.

Bridge

story -> signed change -> table -> v_y -t graph -> $v_y = u_y + a_y t$.

Progression

Fade support deliberately.

Worked

$t=1\text{ s}$, $v_y=12$, $a_y=-10 \rightarrow u_y=22\text{ m/s}$.

Faded

$t=2\text{ s}$, $v_y=6$, $a_y=-9 \rightarrow$ learner completes rearrangement.

Independent

$t=3\text{ s}$, $v_y=3$, $a_y=-9 \rightarrow$ learner chooses model, solves, explains applicability and checks.

Core1A handoff

Teaching authority, not mastery.

Invariant

One clock; signed vertical velocity; constant downward acceleration; apex $v_y=0$.

Boundary

Publication never proves learner mastery.